

Calvin Xiang

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EDUCATION

University of Waterloo

Bachelor of Mathematics

Sept 2022 – Apr 2027

Relevant coursework: Applied Cryptography, Real Analysis, Optimization, Network Flow, Mathematical Statistics, OOP

EXPERIENCE

Member of Technical Staff

SafetyKit

Jan 2026 – May 2025

San Francisco, CA

- Built an autonomous **AI agent platform** using **Claude Computer Use** to automate KYC, gambling-rule, and checkout compliance tests; designed task orchestration and browser automation pipelines.
- Designed a **mass failure recovery system** for a high-throughput ingestion pipeline using **AWS Step Functions**, configurable concurrency backfill tooling, and real-time monitoring for thousands of failed objects.
- Shipped **trust & safety infrastructure** across customer integrations, including **Grindr** spam detection, a **Discord** CSAM hash-check workflow, and **Etsy** listing review systems.

Founding Software Engineer

Ridges AI (Subnet 62, Bittensor Network) 🌐

May 2025 – Dec 2025

Toronto, ON

- Designed core **SWE-agent architecture**, including tool-use flow, prompting strategy, patch generation, and evaluation loops; helped Ridges' agents hit **80% SOTA score on SWE-bench Verified**.
- Created the approval pipeline using **Voyage Code 3** embeddings, **HDBSCAN clustering**, and **t-SNE/UMAP** visualizations to detect code similarity, track agent evolution, and reward only novel solutions.
- Built the secure agent execution infrastructure using **Docker** sandboxes and a **Chutes proxy**, providing isolation, reproducibility, and scalable communication for decentralized **SWE-bench** tasks.
- Scaled Ridges into the second largest subnet, from an early-stage **\$200K** value to a peak **\$100M+ market cap**.
- Led collaboration with a global community of miners, onboarding and supporting **thousands of contributors**, resulting in rapid subnet growth and agent innovation.

Software Engineer

Veryon

Jan 2024 – Apr 2024, Sept 2024 – Dec 2024

San Francisco, CA

- Led the development of a mobile Android app with **.NET MAUI** for offline aviation tracking.
- Integrated **SQLite** for reliable offline data storage, ensuring functionality in low-connectivity environments.
- Significantly reduced operational delays by developing an AI-driven chat tool using **Python** and **Streamlit**, enabling 300+ engineers to efficiently edit 10,000+ maintenance documents from **AWS S3**.
- Optimized document retrieval by **70%** with a PDF sourcing feature using **AWS Kendra** for faster access.
- Added a text-highlighting feature using **LangChain NLP** for faster navigation of technical documents.

Software Engineer

Matrix Leap

Jan 2024 – Apr 2024

San Francisco, CA

- Designed calling agents using **Grok**, currently utilized by **Bland.ai**, to automate over **200** business workflows with voice-to-action capabilities.
- Built a coding platform that seamlessly integrated the **Google Sheets API** to source questions, solutions, and metadata, paired with **OpenAI LLMs** to provide personalized, real-time feedback for users.
- Engineered a scalable **FastAPI**-based microservice for automated grading of coding exercises, capable of processing up to **1,000** submissions per minute while consistently maintaining a **99.8%** accuracy rate.

Software Developer Jr.

Equifax Canada Inc.

May 2023 – Aug 2023

Toronto, ON

- Built scalable **CI/CD pipelines** with **Jenkins**, improving deployment efficiency and reducing release cycles.
- Helped with the migration **100+ TB** of client data from **IBM AS/400 Servers** to **GCP**.
- Automated data transfers using optimized **bash scripts**, reducing processing time.

TECHNICAL

Languages: Python, TypeScript, JavaScript, Java, C#, C/C++, Go, SQL, Dart, HTML/CSS

Backend & Cloud: FastAPI, Node.js, AWS CDK, **Lambda**, **Step Functions**, Docker, Kubernetes, GCP, Jenkins

AI/ML: OpenAI, Claude, **Claude Computer Use**, LangChain, Hugging Face, Llama 3, **HDBSCAN**, t-SNE, UMAP

Frontend & Mobile: React, Next.js, Streamlit, .NET MAUI

Datastores & Search: PostgreSQL, SQLite, Redis, Supabase, Pinecone, **Turbopuffer**, AWS Kendra, pgvector

CERTIFICATIONS

Oracle Certified Java Programmer

Carnegie Mellon picoCTF Second Place

Sept 2023

Mar 2022